清華大學地球系统科学系
Department of Earth System Science, Tsinghua University

Biogeochemical and biophysical effects of tropical forest disturbance

Lei Zhu (l-zhu20@mails.tsinghua.edu.cn)

Wei Li, Philippe Ciais, Zhe Zhao, Yidi Xu

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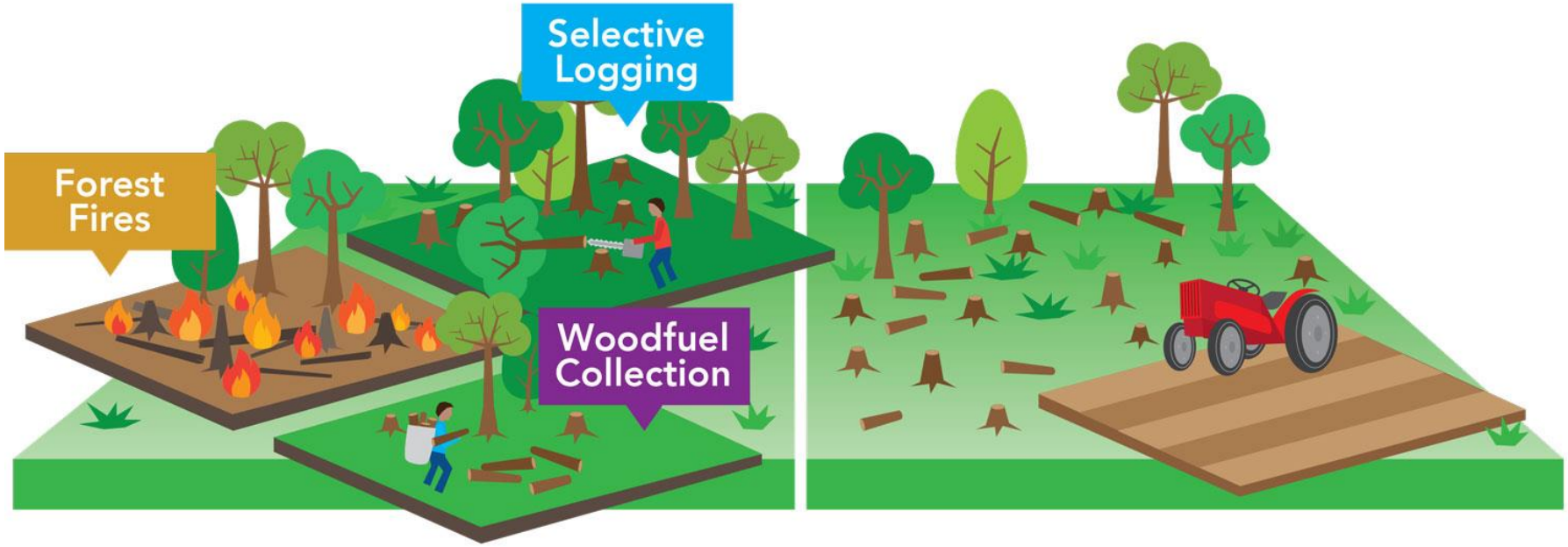
Forest disturbance：deforestation + forest degradation

- Deforestation：
  - Land use changed
  - Happens on a large scale
- Forest degradation：
  - Land use unchanged
  - Happens on a small scale

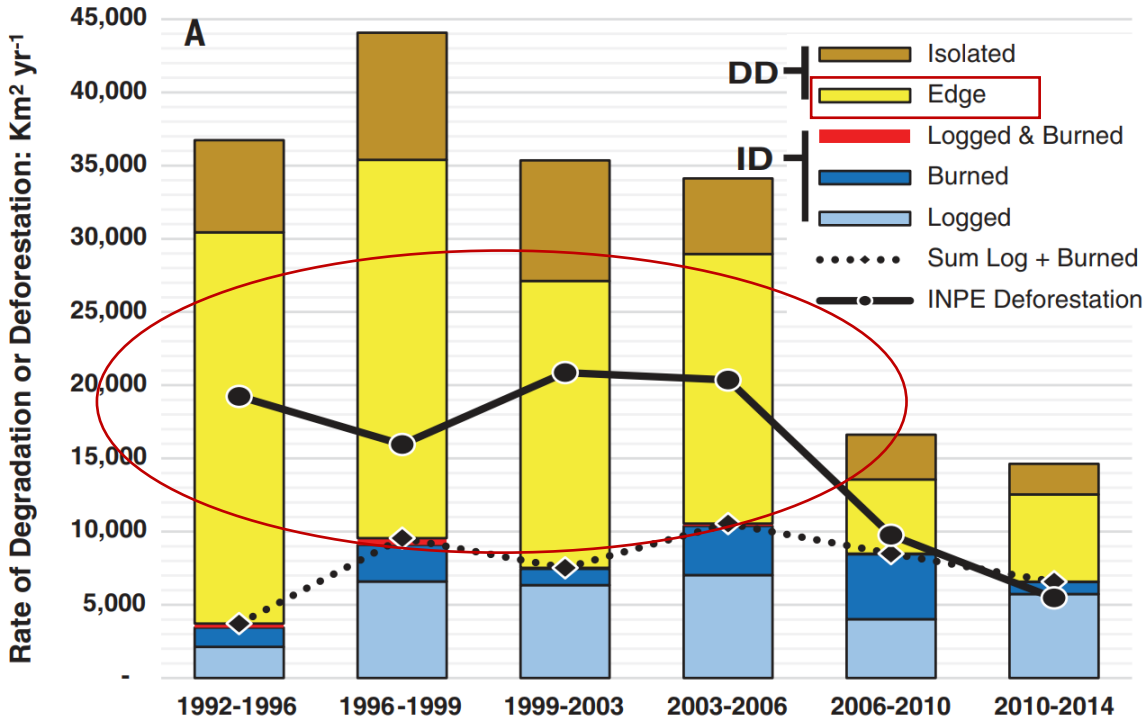


Degradation vs. Deforestation

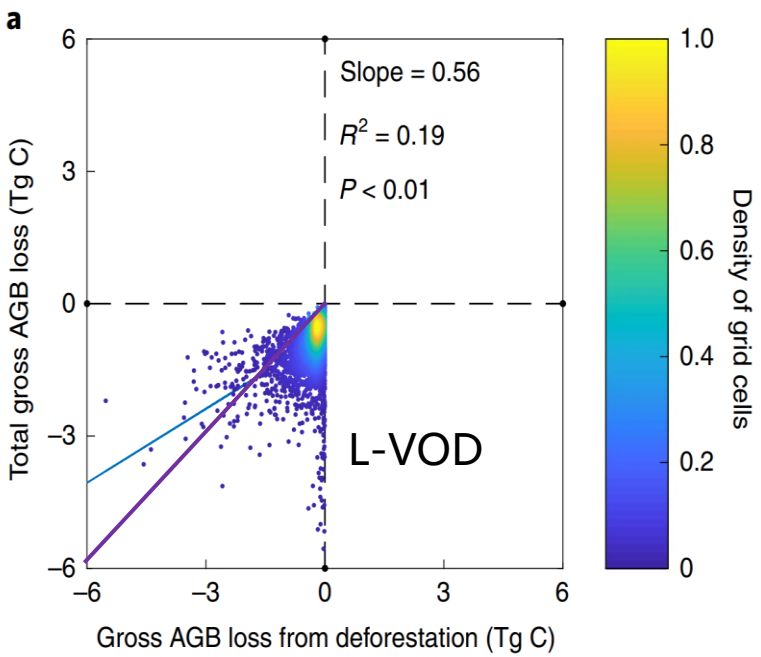
winrock.org

Forest degradation

- Area
  - Amazon forest：degradation>deforestation
- Carbon loss
  - Remote sensing：forest degradation（73%）>deforestation（27%）



Matricardi et al., 2020



Qin et al., 2021

Edge forest

- Mechanism：Wind, drought, fire…
- Field obs：8.8±10.2% biomass loss in the first 10 to 17 years after fragmentation

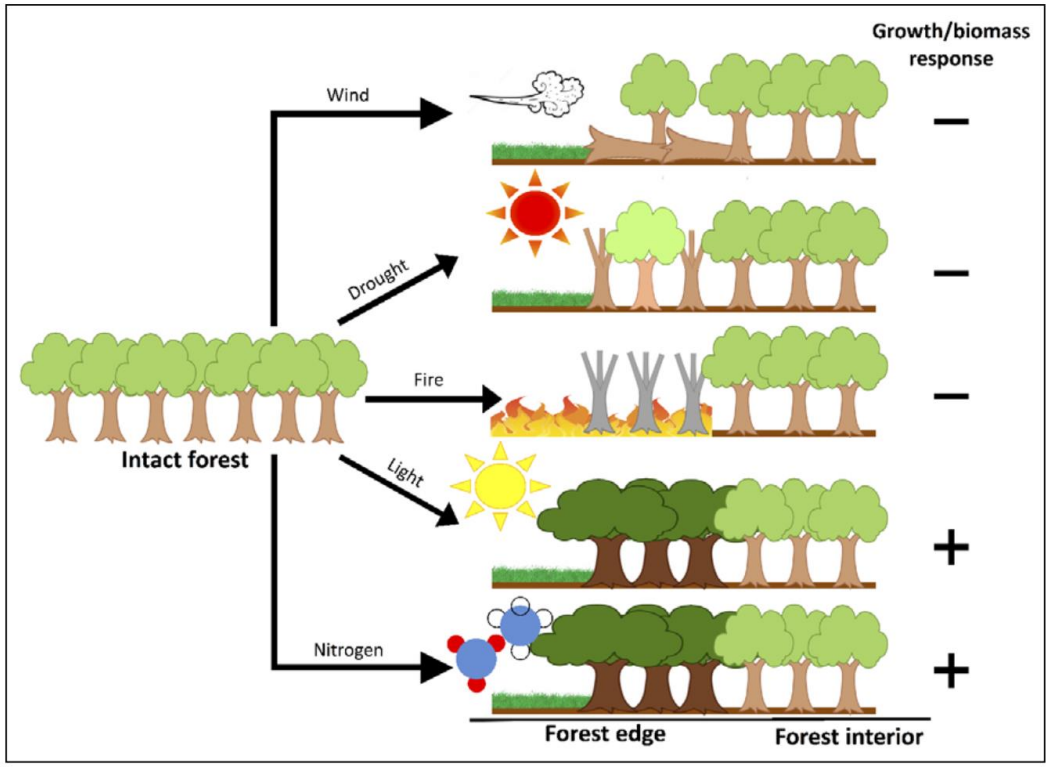
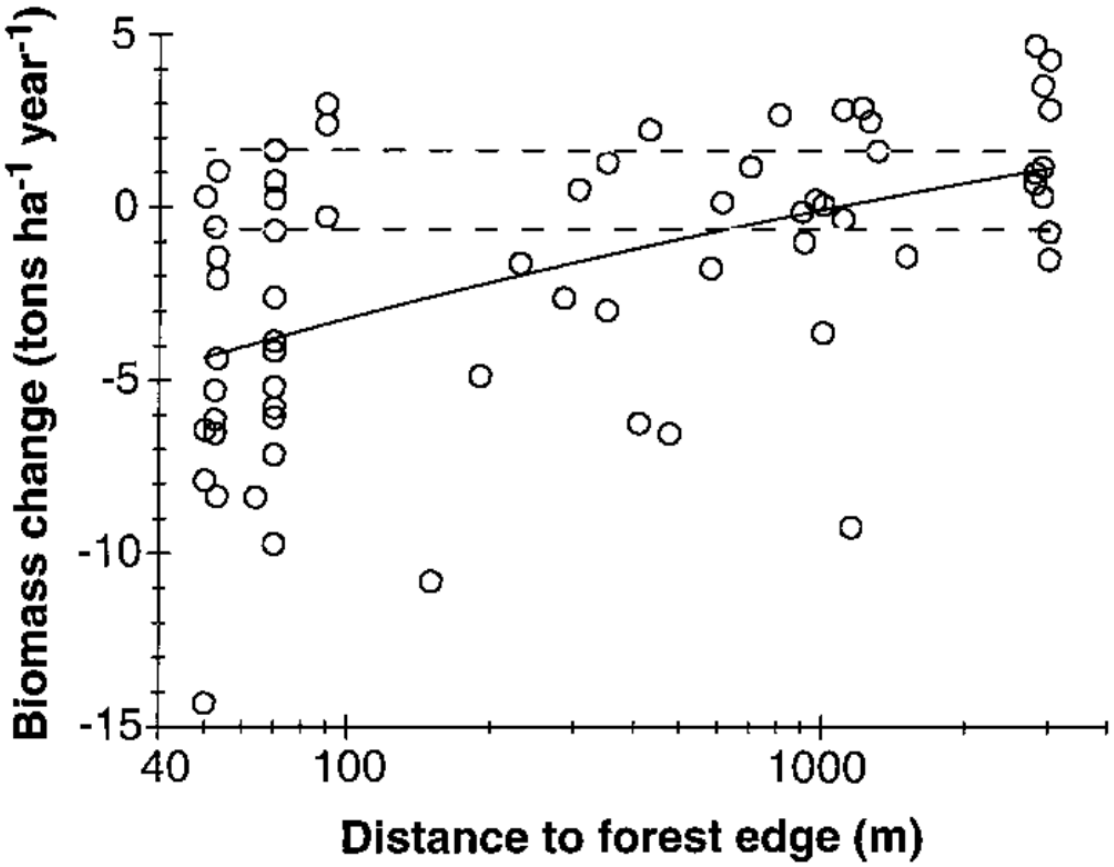


Figure 3. Changes in the forest edge microenvironment include exposure to more wind, drought, fire, light, and nitrogen. The positive (+) sign and negative (–) sign indicate favorable and adverse conditions, respectively, for growth/biomass.

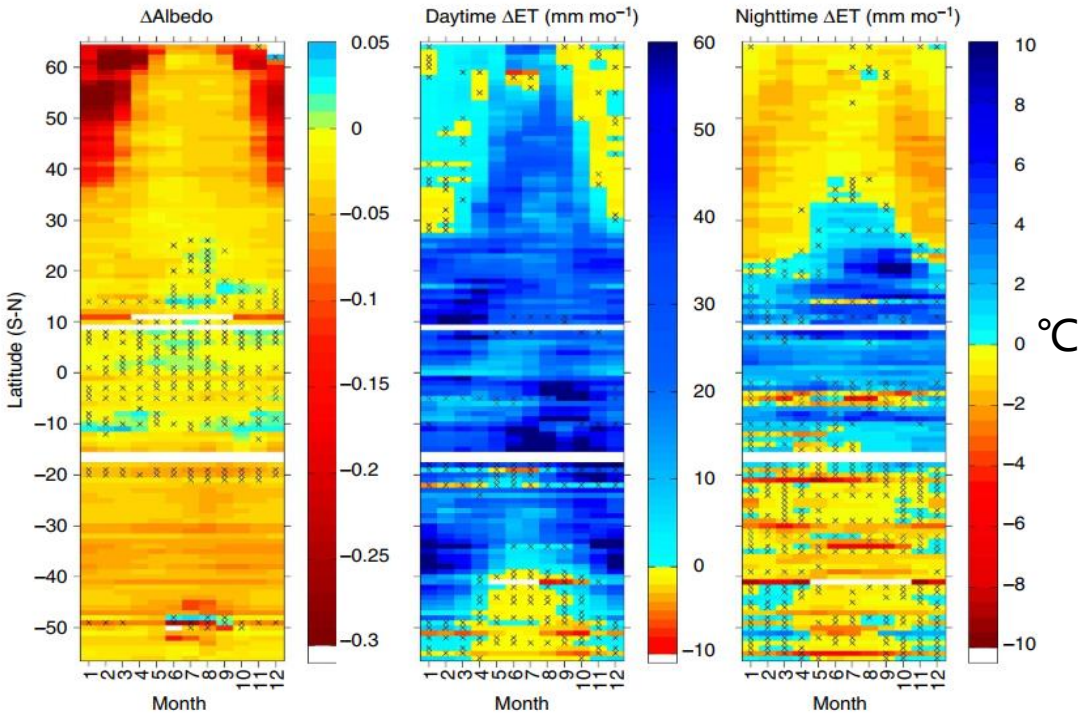
Smith et al., 2018

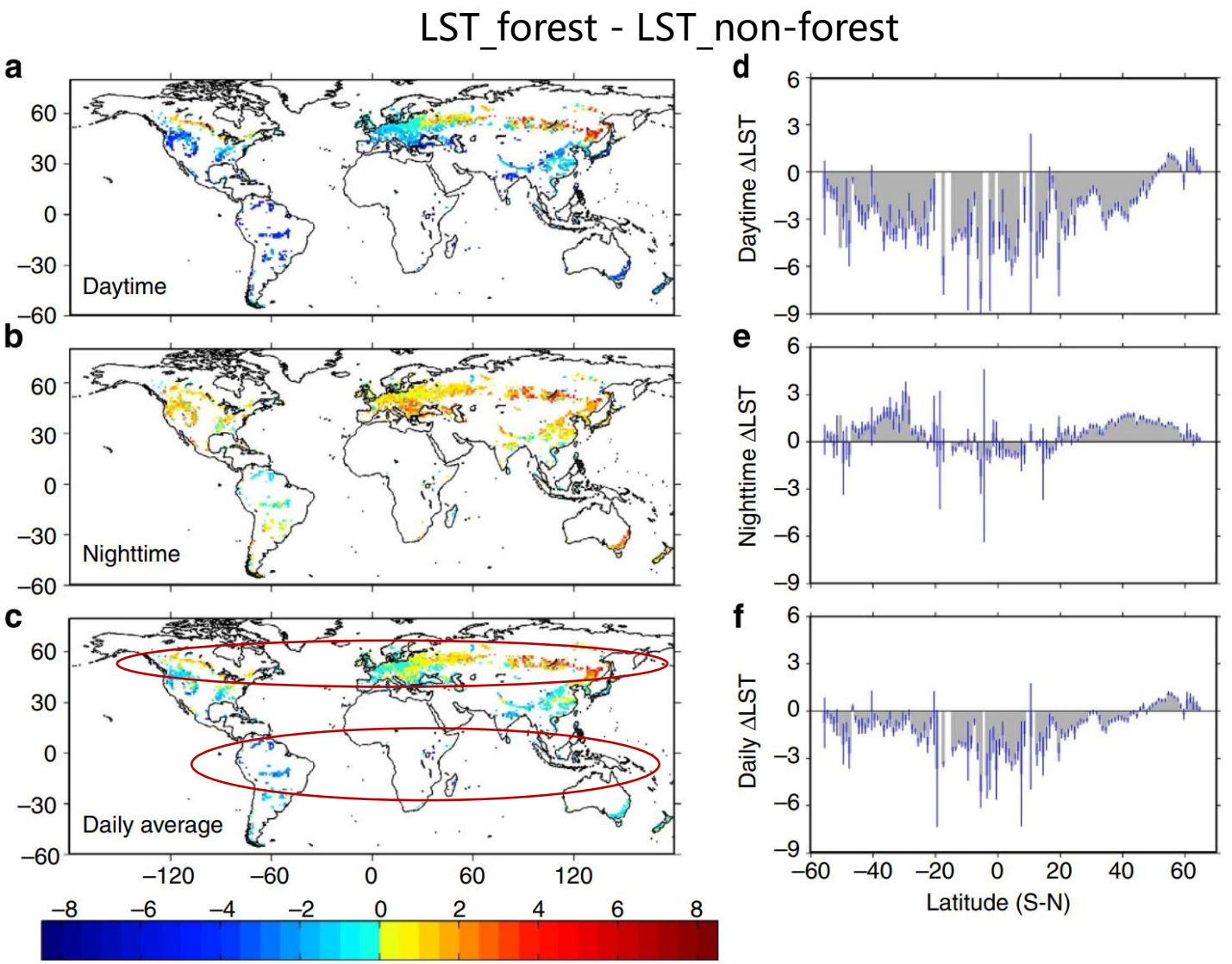


Laurance et al., 1997

Biophysical effects

- Cooling effects in the tropical forest
  - Higher transpiration
  - Higher roughness
- Warming effects in the boreal forest
  - Lower albedo





Li et al., 2015

Research gap

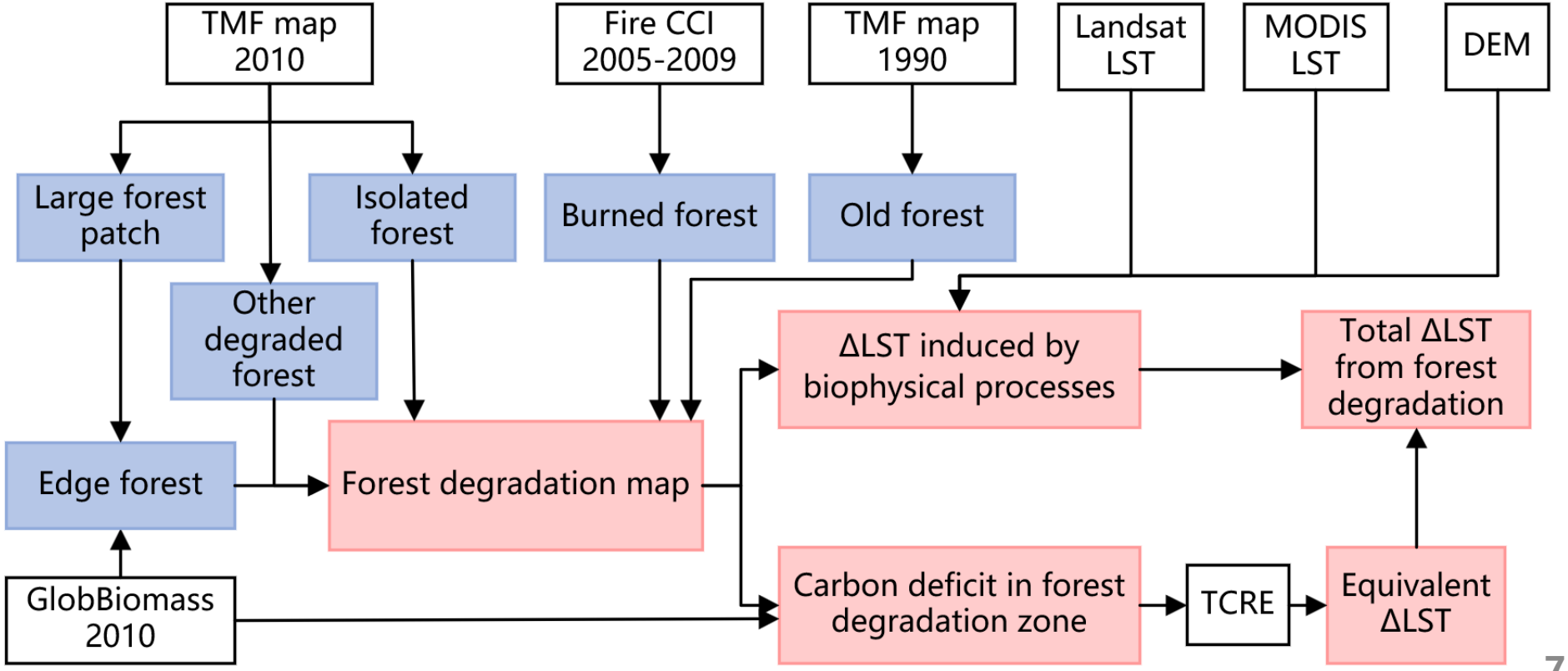
1. How much area of the tropical rainforest has been degraded?
2. What are the biogeochemical and biophysical effects of tropical forest degradation?
3. What’ s the impact of multiple disturbances in tropical forests?

Method

Data:

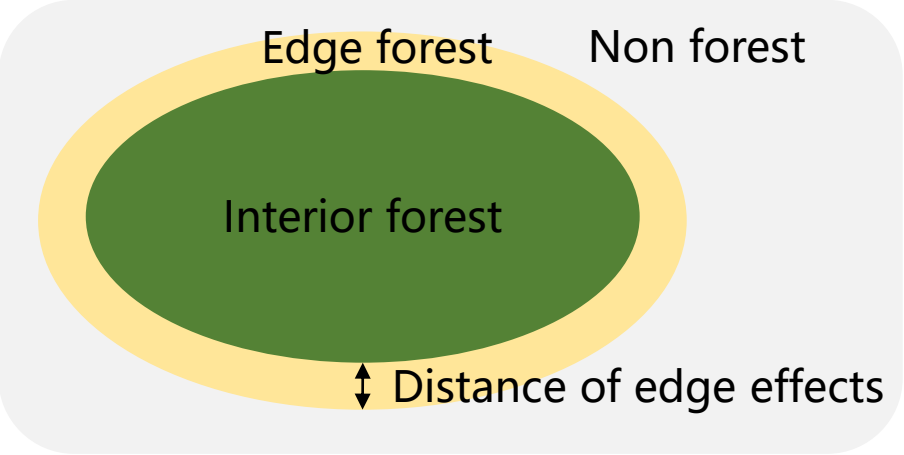
- Tropical moist forest（TMF，30 m）
- Biomass map（GlobBiomass，25 m）
- Fire map（Fire CCI，250 m）
- Land surface temperature（Landsat，30 m; MODIS, 1 km）

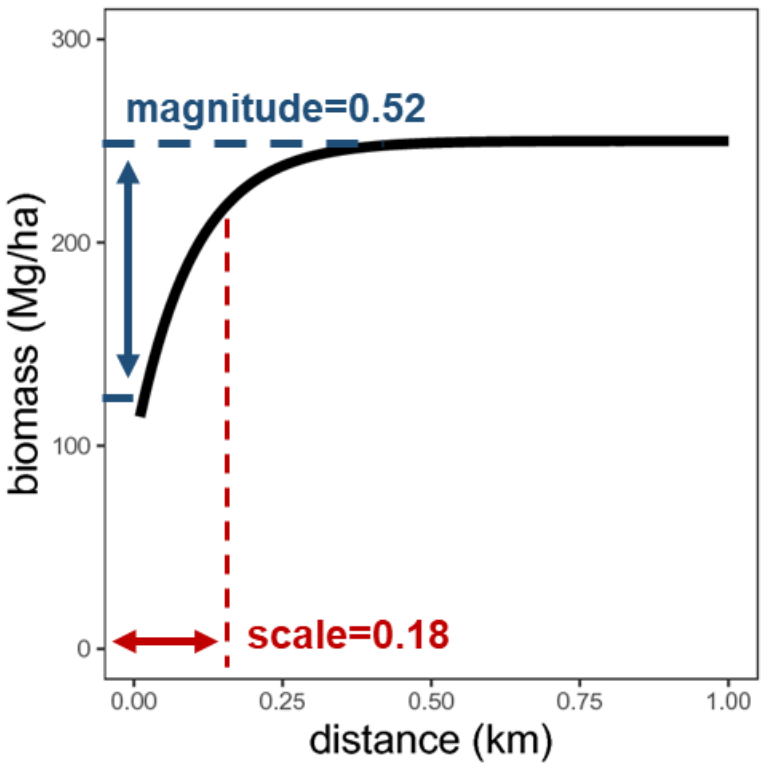
Space for time

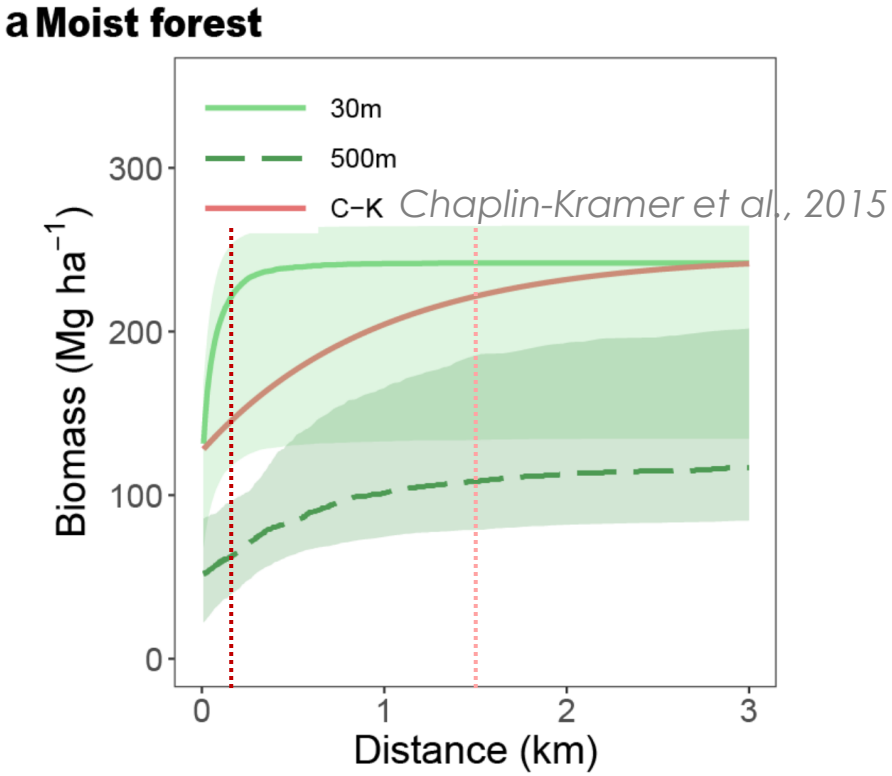


The distance of edge effects from remote sensing

- Remote sensing:
  - Resolution 500 m: ~1.5 km
  - Resolution 30 m: 0.15~0.21 km
- Field obs: within 0.3 km





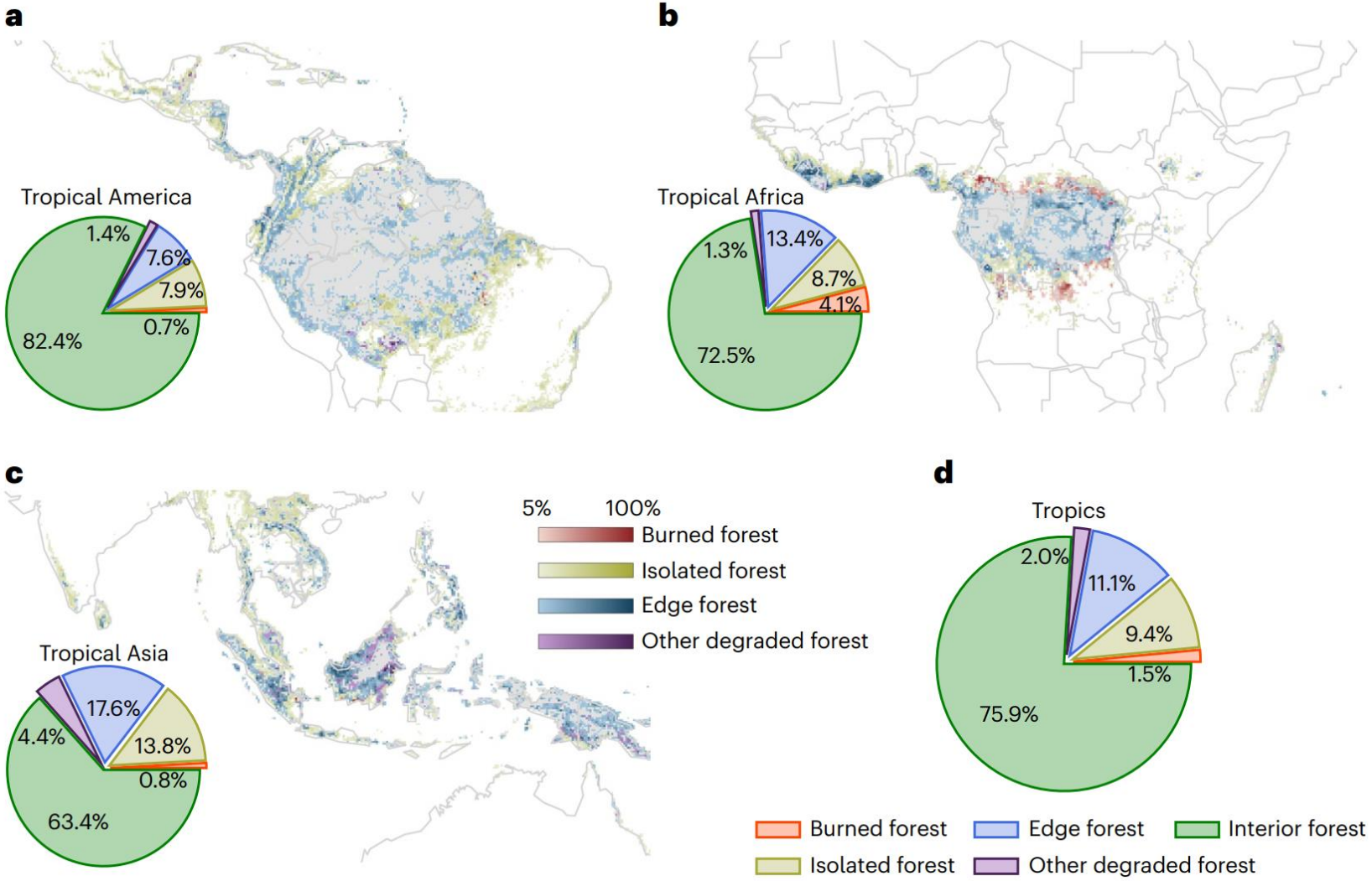


Forest degradation map

- About ¼ of the total moist forest was degraded in 2010
- Dominated by edge forest and isolated forest

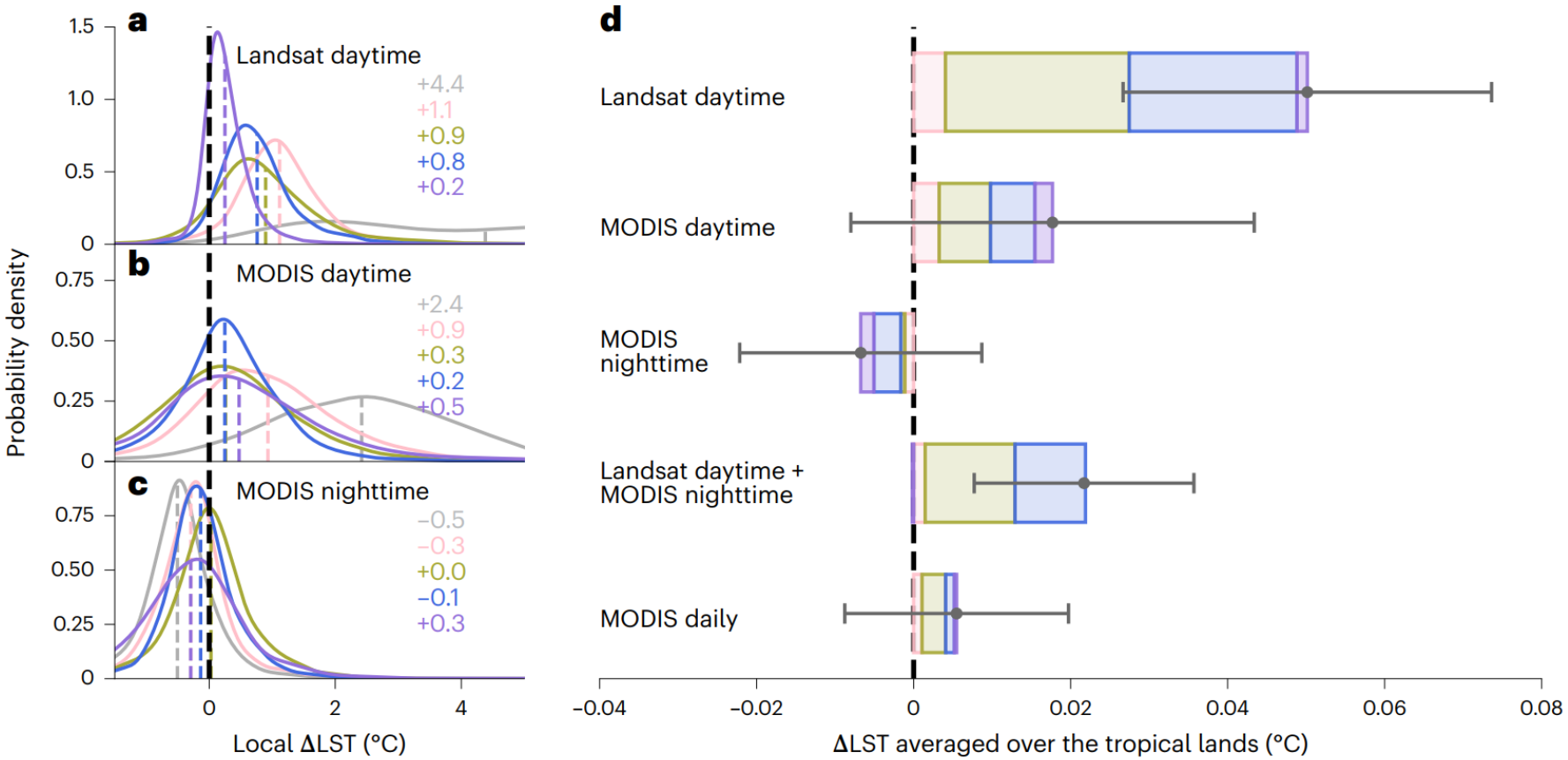
Four types of degraded forest:

1. Burned forest
2. Isolated forest
3. Edge forest
4. Other degraded forest (wood harvest、extreme weather)



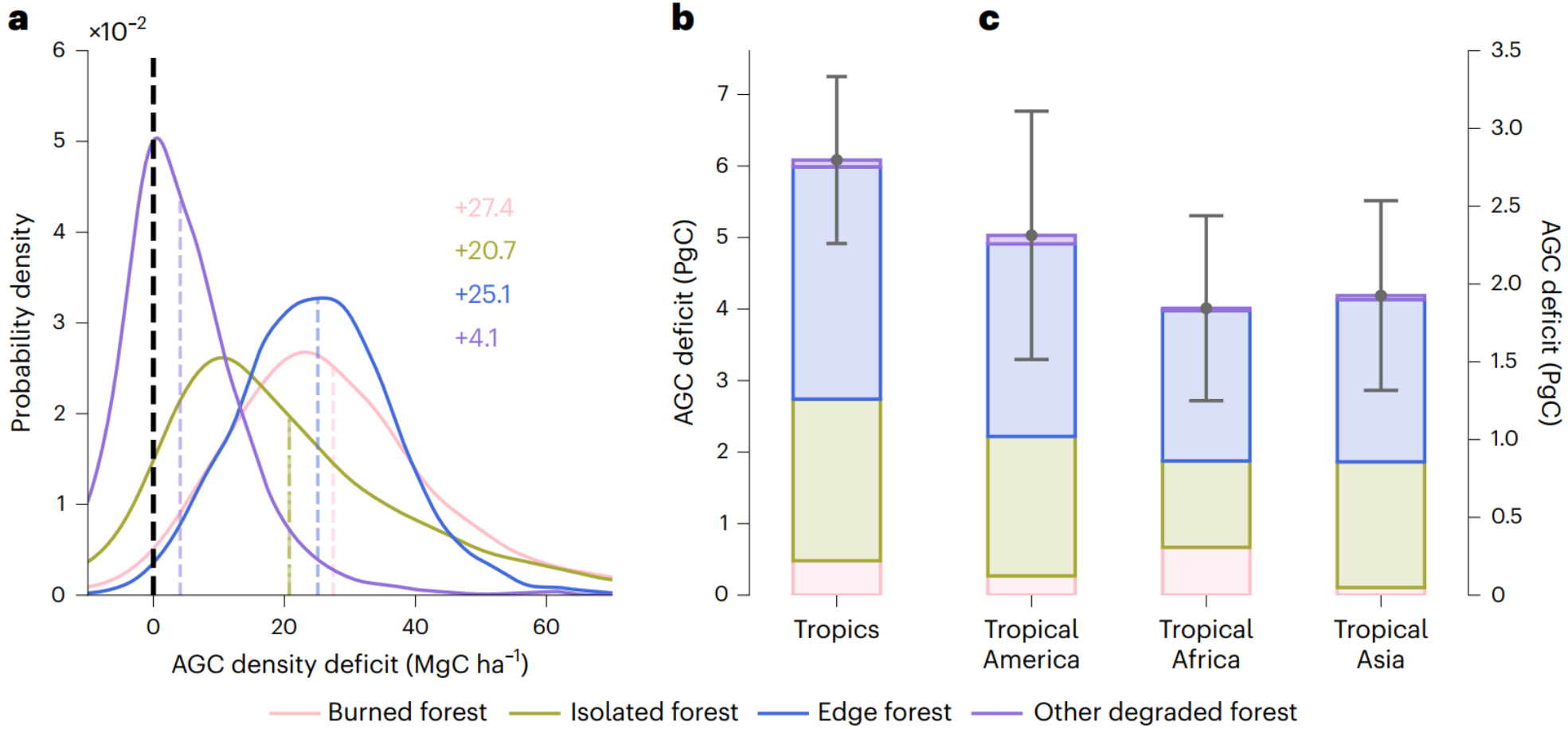
Biophysical effects of forest degradation

- LST_degraded-LST_intact=0.78±0.88 °C
- Similar signal from both Landsat LST and MODIS LST data
- Warming effects for the whole tropical lands： 0.022±0.014 °C



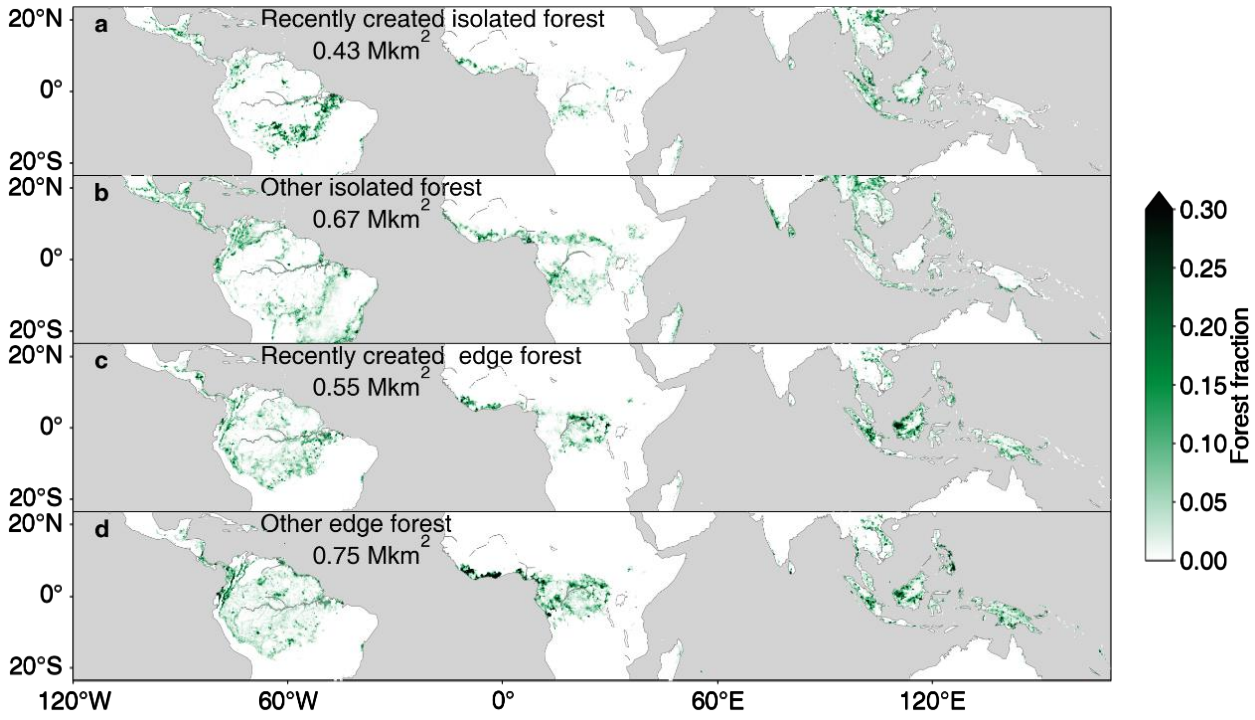
Biogeochemical effects of forest degradation

- Carbon density (degraded)-Carbon density (intact)=21.8±16.6 MgC ha^{-1}
- Carbon deficit：6.1±2.0 PgC
- Equivalent warming：0.026±0.013 °C
- The magnitude of BGC warming（0.026 °C）and BPH warming（0.022 °C）is comparable

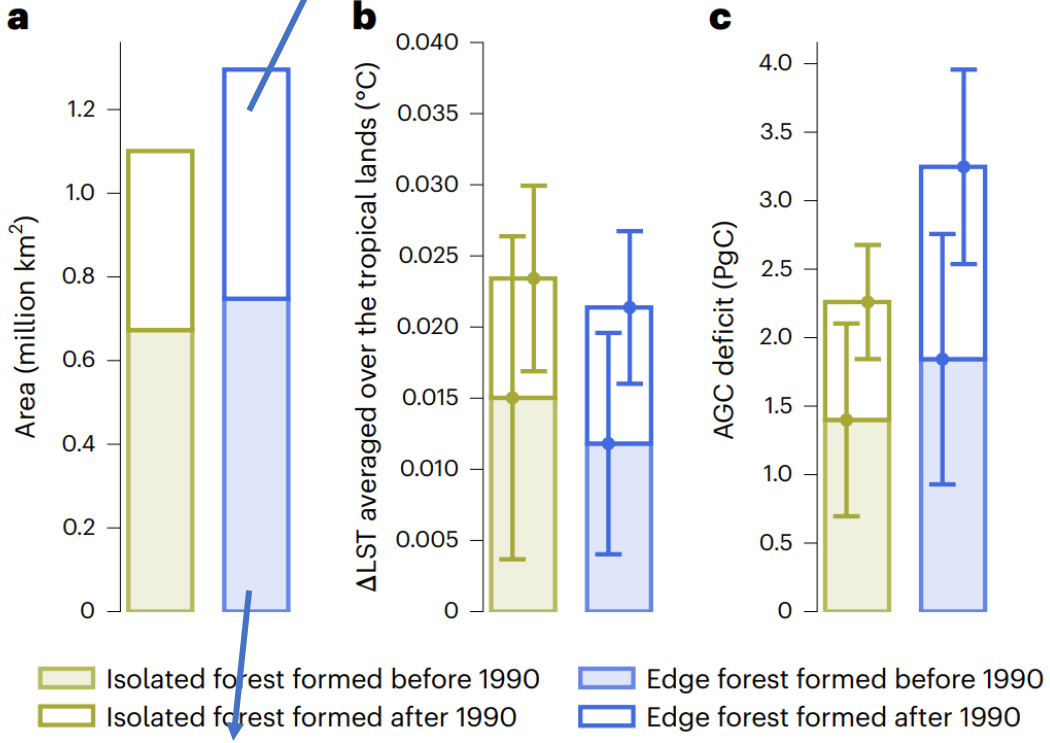


Degradation from recent anthropogenic disturbances

- More than a third of degraded forests have been created since 1990
- Daytime warming effects: 0.018±0.008 °C
- Carbon deficit: 2.3±0.8 PgC



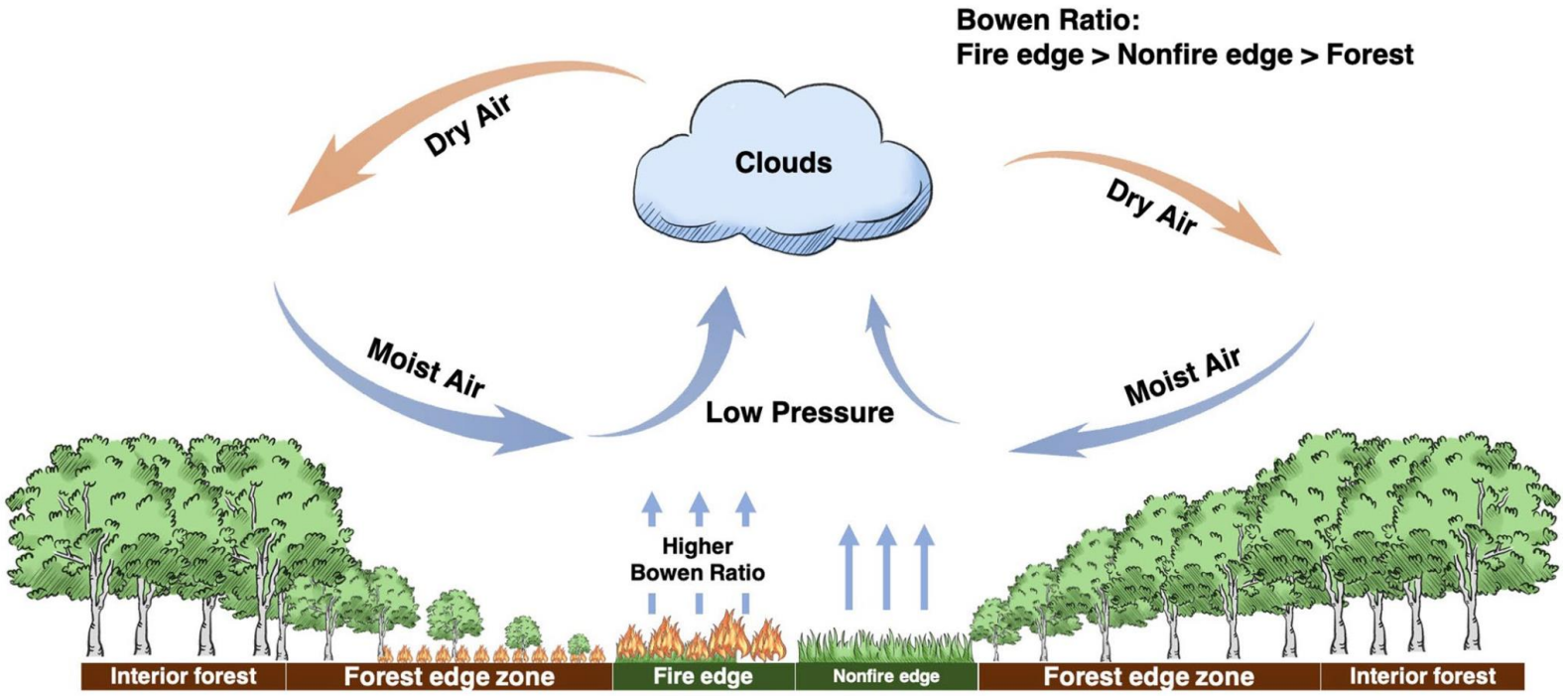
Newly created after 1990



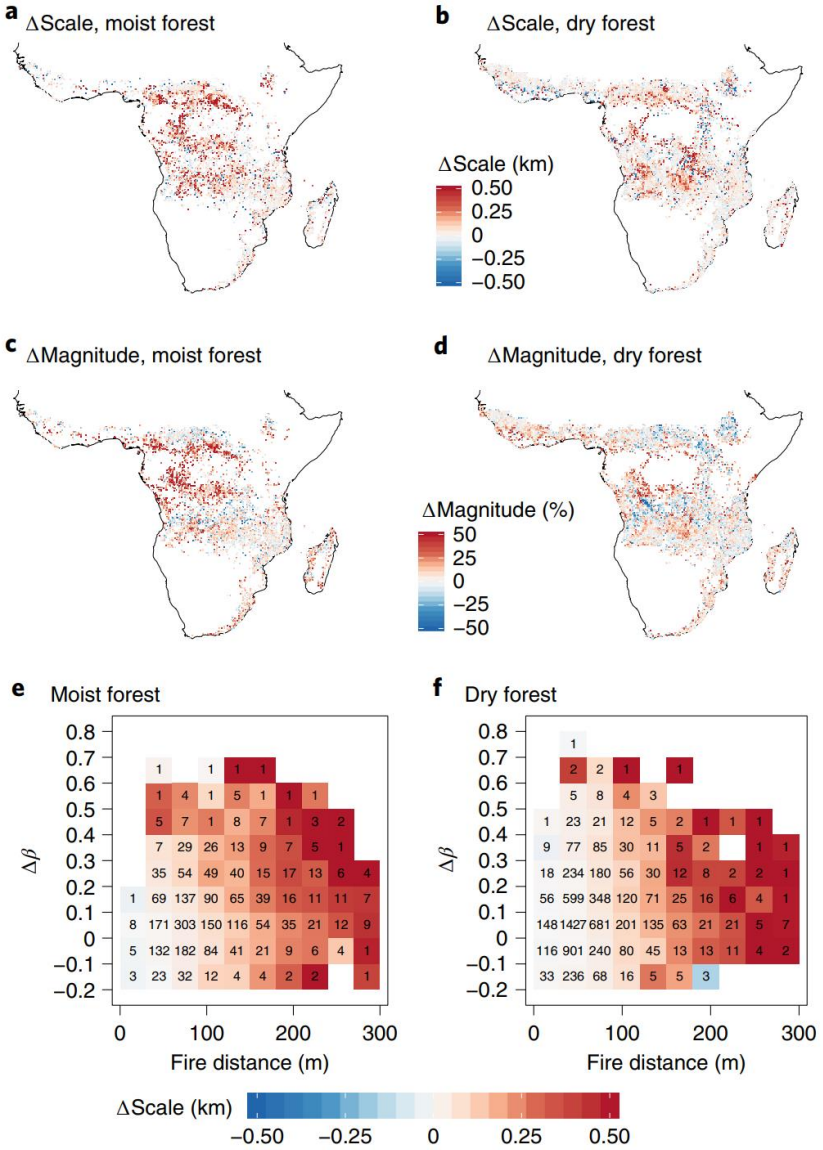
Exist before 1990

Fire impacts on edge effect

- Higher carbon deficit and edge distance in fire edge
- Fire enhances edge effects
  - Direct: intrude into forest
  - Indirect: change in local atmospheric circulations and canopy dryness

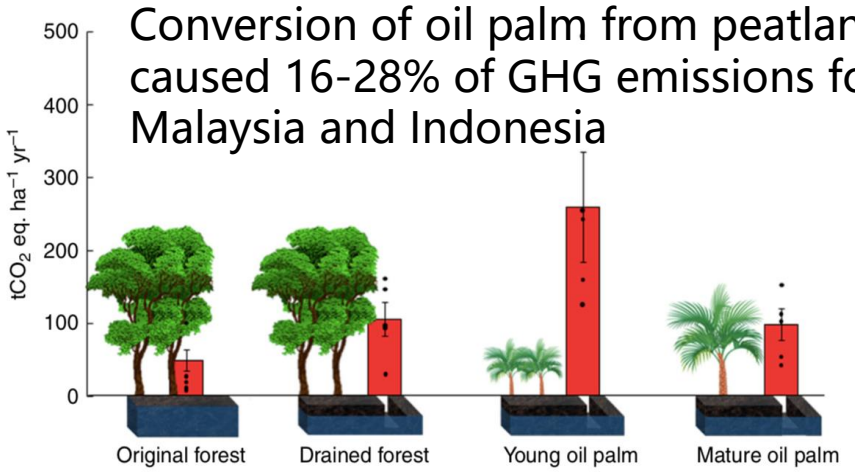


Fire edge-Nonfire edge



Deforestation from oil palm

- Indonesia and Malaysia consist of 65% of the world’ s oil palm plantation
- A 6-fold increase in oil palm plantation area during the past 40 years（FAOSTAT, 2020）
- Oil palm expansion causes deforestation, declination in biodiversity, loss of peatland and greenhouse gas emission

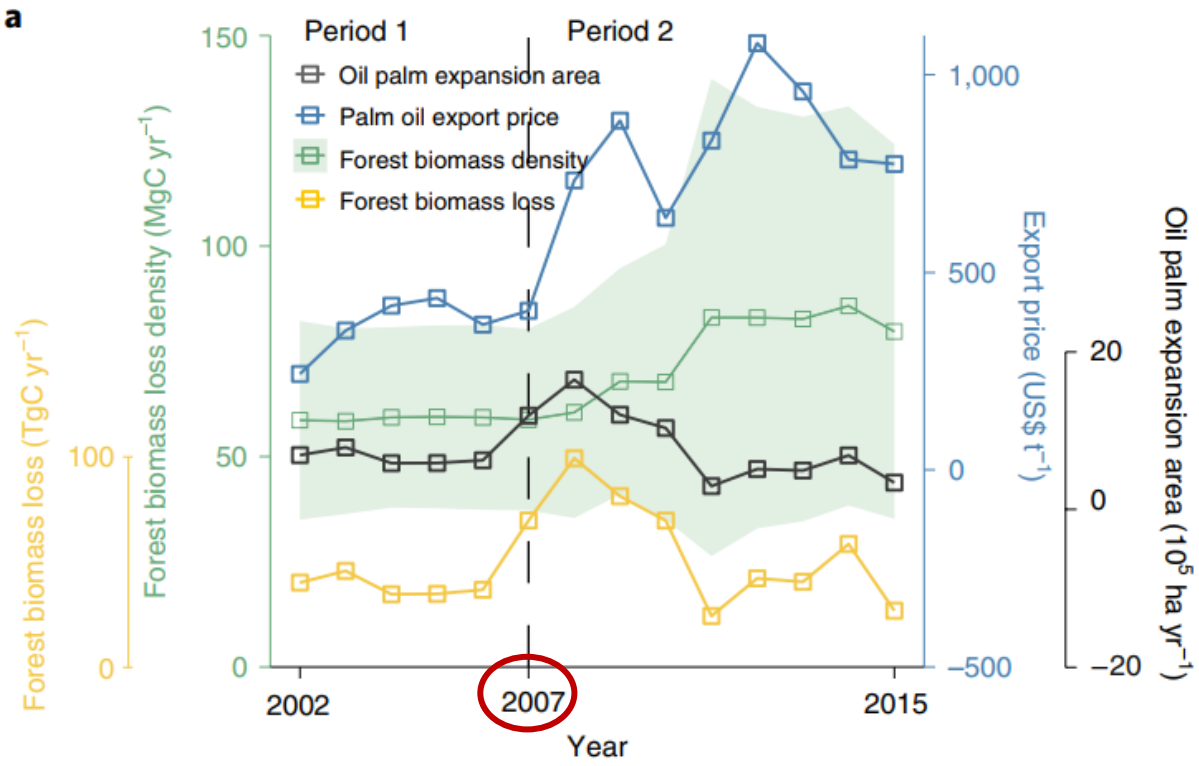


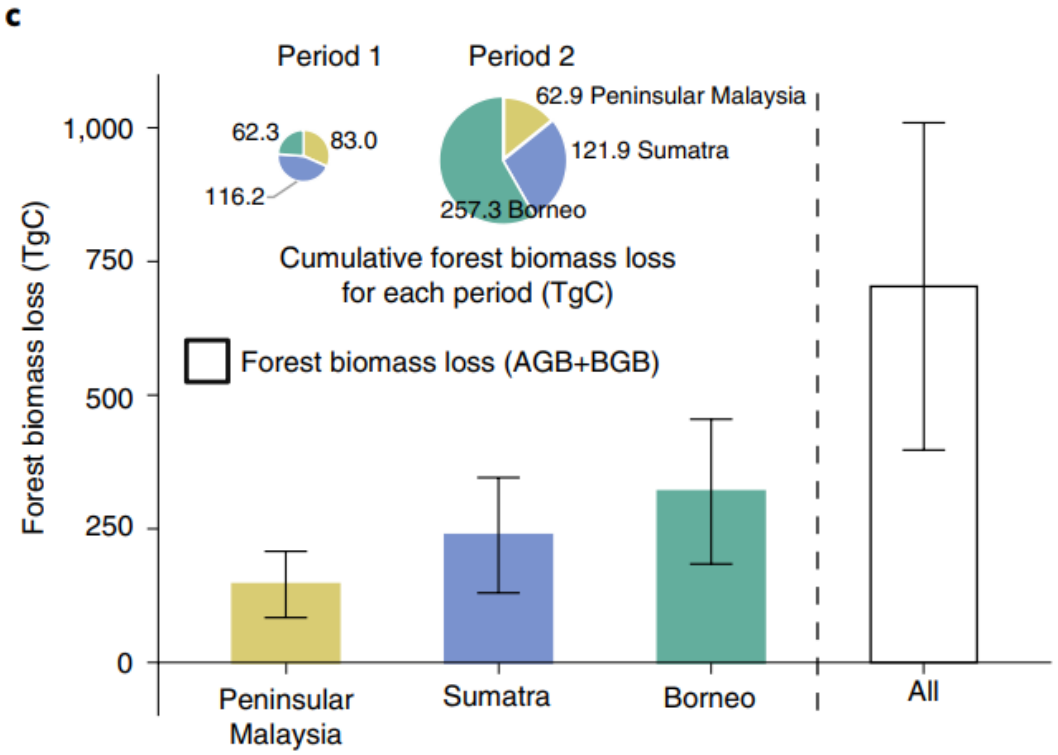
(Cooper et al., 2020)

It is essential to accurately quantify the spatial-temporal pattern of oil palm expansion in SE Asia

Carbon loss from oil palm expansion

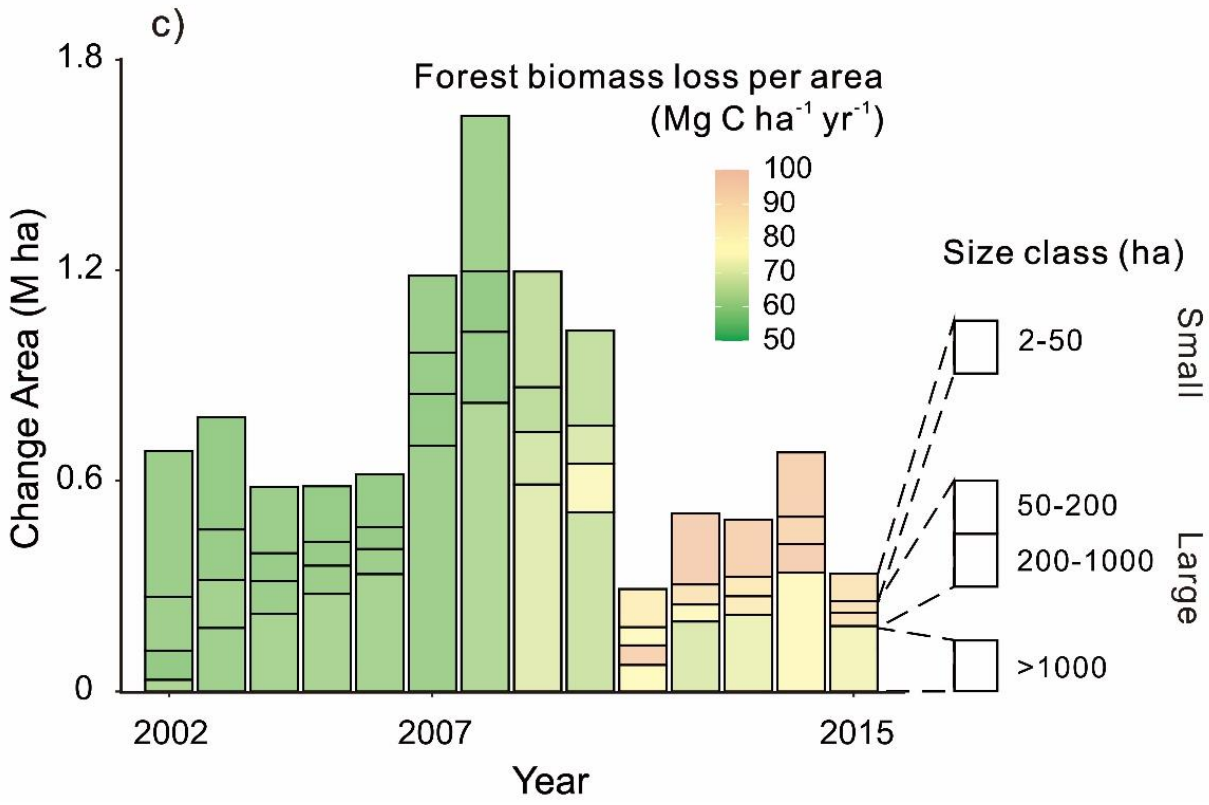
- Overlap: oil palm maps + GlobBiomass map + Hansen’ s tree loss maps
- Biomass loss during 2001-2015: 50 TgC/yr
- Expanded into forests with higher carbon density after 2007

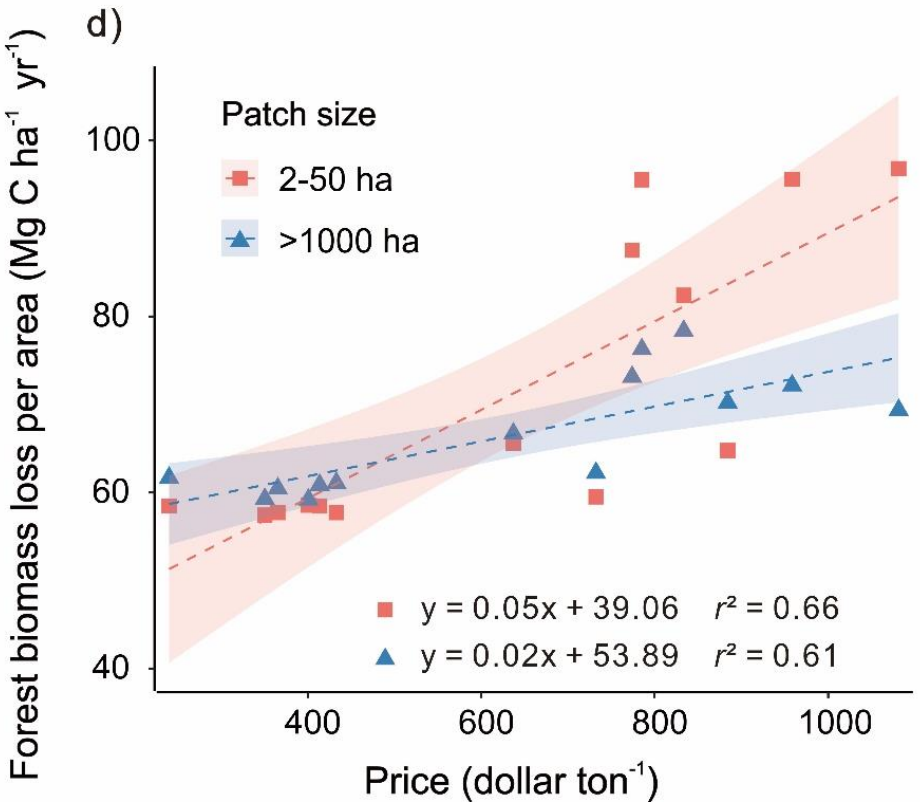




Oil palm expansion from different patch sizes

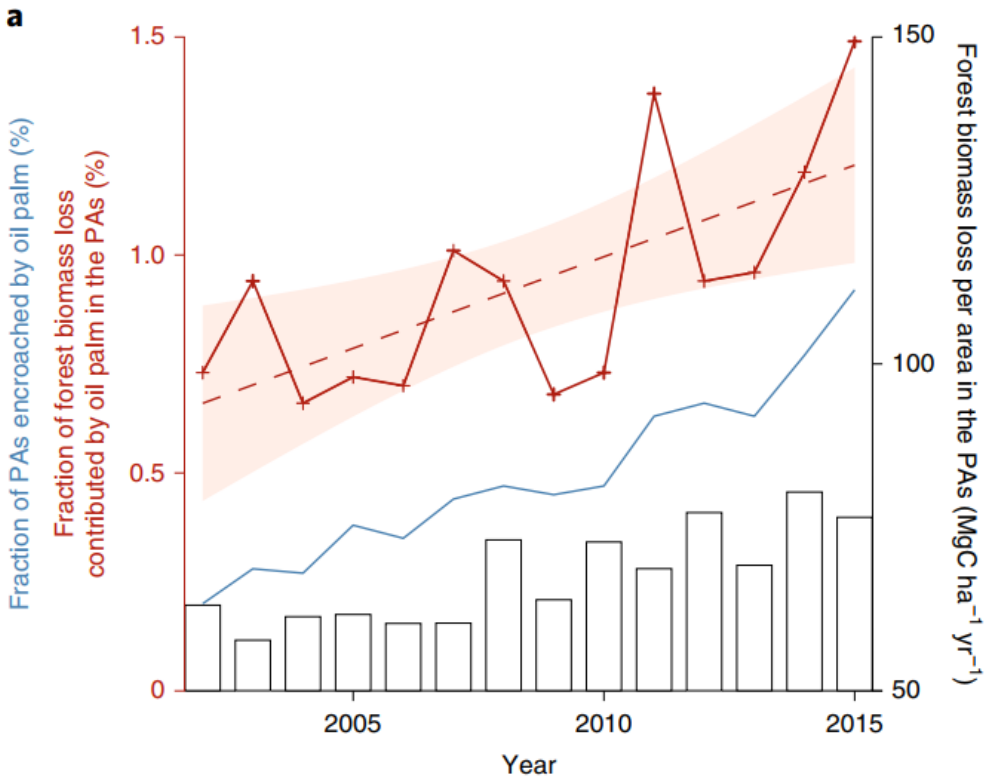
- The small-scale oil palm plantations have expanded from low to high biomass density forest
- Higher export prices may accelerate small-scale plantation expansion into areas with higher yield potential.

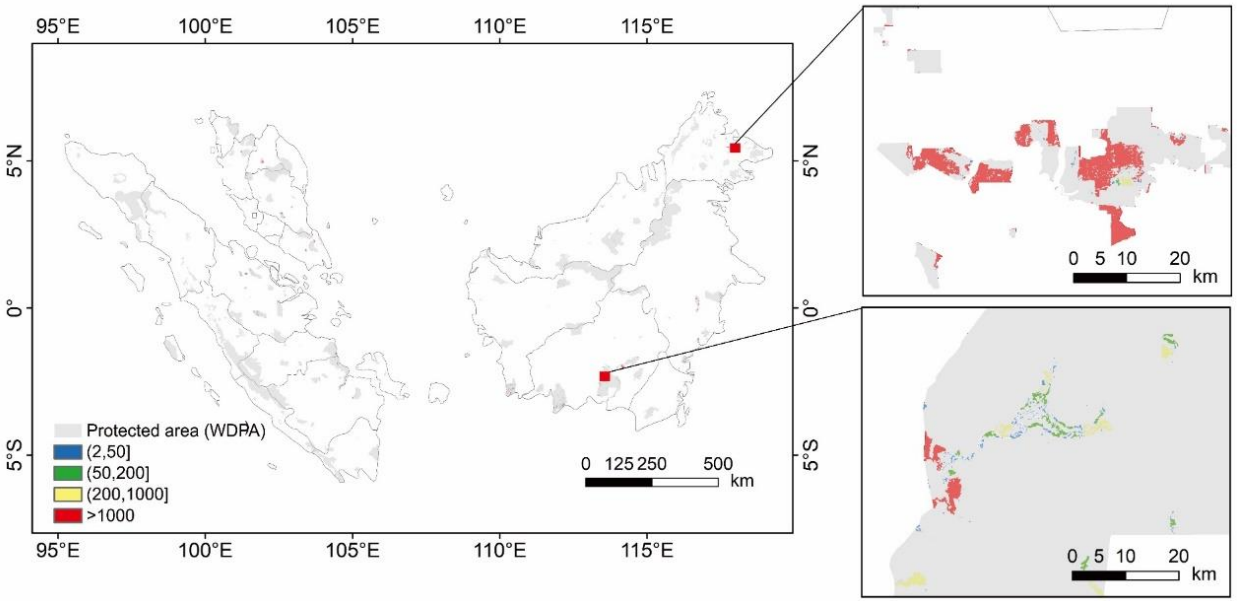




Oil palm expansion in protected area

- Overlap: Protected area (WDPA) + oil palm
- Carbon loss caused by oil palm within protected areas increased faster than regions outside protected areas.





Overlapped oil palm plantations and protected areas from WDPA

Conclusions

- The biogeochemical and biophysical warming effects of tropical rainforest degradation are comparable.
- fires exacerbate the edge effects of forests through both direct and indirect impacts
- The encroachment of oil palm plantations from low to high biomass density forests is particularly obvious for small-scale plantations after 2007.

Reference：

1. *Zhao, Z. et al. Fire enhances forest degradation within forest edge zones in Africa. Nature Geoscience, doi:10.1038/s41561-021-00763-8 (2021).*
2. *Zhu, L. et al. Comparable biophysical and biogeochemical feedbacks on warming from tropical moist forest degradation. Nature Geoscience, doi: 10.1038/s41561-023-01137-y (2023).*
3. *Xu, Y. et al. Recent expansion of oil palm plantations into carbon-rich forests. Nature sustainability, doi: 10.1038/s41893-022-00872-1 (2022).*

Thanks!

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